LightGBM for Oral versus Non-Oral Classification of Head and Neck Squamous Cell Carcinoma from Multi-Omic TCGA Data: A Replication on an Independent Pipeline

Technical report

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Abstract

We evaluate LightGBM [4], a gradient-boosted decision tree ensemble, on the oral-versus-non-oral head and neck squamous cell carcinoma classification task from The Cancer Genome Atlas, using an independently built data-loading and preprocessing pipeline (leak-free per-fold feature selection, 5-fold stratified cross-validation, three of the six molecular data types used in prior work on this task: gene expression, DNA methylation and microRNA). Two settings are compared: a single point-estimate LightGBM model per data type and combination, and a 30-member bootstrap ensemble giving a per-patient predictive uncertainty score. DNA methylation is the strongest single data type in both settings (accuracy 0.927 point-estimate, 0.929 bootstrap-ensemble), and combining all three data types makes no measurable difference relative to methylation alone (0.931 point-estimate, 0.927 ensemble). Bootstrap ensembling changes accuracy by between -0.015 and $+0.006$ with no consistent direction, and adds a per-patient uncertainty estimate that tracks data-type reliability: among the single data types, methylation carries the lowest mean predictive uncertainty (0.0365) and microRNA the highest (0.0844). Mapping the top methylation probes to their nearest genes points to a specific, biologically coherent explanation for methylation's dominance: the recurring genes (including *PITX2* and multiple *HOX* cluster members) are established regulators of pharyngeal arch and craniofacial development, consistent with methylation retaining a signature of the oral cavity's distinct embryonic origin. An extended study runs LightGBM on all six data types and all 63 combinations of them: methylation stays the strongest single type, and no combination differs meaningfully from the better of its own members (55 of 57 comparisons not significant, the other two about what chance predicts). Both LightGBM settings are compared against an independently reproduced, corrected implementation of the MoXGATE multi-omic attention network evaluated on the identical patient cohort and cross-validation folds.

1 Introduction

A companion pair of reports on this same task – an improved implementation of the MoXGATE multi-omic attention network, and a LightGBM baseline evaluated exhaustively across every combination of six data types – found that gradient-boosted trees matched or modestly exceeded a corrected multi-omic neural network on DNA methylation, and did so with substantially less sensitivity to which and how many data types were combined. This report evaluates LightGBM independently, on a separately built data pipeline, to check whether that pattern reproduces under a different (but overlapping) implementation, and extends the comparison with an uncertainty-aware ensemble setting the original LightGBM report did not include. It also repeats that report's study of all 63 combinations of six data types on this pipeline.

2 Data

2.1 Source and cohort

Data come from the TCGA-HNSC project [7], accessed via UCSC Xena [8] (both the TCGA Hub and the GDC Xena Hub). The aligned cohort used throughout this report has 477 patients with all three data types present and a valid oral/non-oral label, of whom 61% are labelled oral. The combination study (Section 4.3) uses all six data types on smaller, common patient groups described in Section 3.5.

2.2 Data types

Three data types (gene expression, methylation, microRNA) are used in the main comparison, matching the three-modality scope of MoXGATE's architecture. All six are used in the combination study.

Data type	Source	Features before preprocessing
Gene expression (RNA)	TCGA Hub, HiSeqV2, log-normalized	20,531 genes
DNA methylation	TCGA Hub, HumanMethylation450, beta values in $[0, 1]$	485,578 probes
microRNA	TCGA Hub, miRNA_HiSeq, log-normalized	2,247
Copy number	TCGA Hub, GISTIC2 thresholded gene-level calls	24,777 genes
Somatic mutations	GDC, somatic mutation (WXS); 1 if any mutation is recorded in the gene, silent variants included	16,509 genes
Protein (RPPA)	GDC, protein	488 antibody targets

These are the same tumours as in the companion report but not identical data releases. The companion used the GDC release for all six types (for example ASCAT3 integer copy number for 60,623 genes, non-silent mutations only, 1,881 microRNA precursors, 60,660 expression genes). Here copy number is GISTIC2 thresholded calls, and the mutation matrix marks a gene as mutated if any mutation is recorded in it, silent or not. Results are therefore comparable in pattern with the companion report, not to the third decimal.

2.3 Preprocessing

Each data type is preprocessed separately within each cross-validation fold, fitted on the training patients of that fold only: features missing in more than 40% of training patients are dropped, remaining gaps are filled with the training median, and the top 5,000 features by one-way ANOVA F-statistic against the training labels are kept (fewer-feature data types, such as microRNA, keep everything). This matches the preprocessing used for the companion MoXGATE report's own evaluation on the same pipeline, and every LightGBM and MoXGATE number reported here was produced with exactly this preprocessing. Methylation's raw file (485,578 probes) is narrowed to 20,000 candidates by the same ANOVA F-statistic, computed in a memory-bounded streaming pass over the raw file, before the per-fold 5,000-feature selection is applied – a practical necessity for fitting the full probe set in memory, not a step present in the companion reports' own pipelines.

3 Method

3.1 Evaluation

All results use 5-fold stratified cross-validation on the 477-patient cohort, with the same fold assignments shared across every data type and model setting reported here, and across the MoXGATE comparison figures in Section 6.

3.2 LightGBM configuration

Parameter	Value
Number of trees	500
Learning rate	0.05
Number of leaves	31 (leaf-wise growth)
Feature fraction per tree	0.3
Row (bagging) fraction	0.8, resampled every iteration
Minimum samples per leaf	20
Minimum child weight	10^{-3}
L1 / L2 regularization	0 / 0

This configuration matches the LightGBM report referenced in the Introduction, to keep the two evaluations as comparable as the difference in underlying data pipeline allows.

3.3 Point-estimate setting

One LightGBM model per data type (and one on all three data types concatenated, routed through the same per-type preprocessing via a column transformer) is trained and evaluated with the cross-validation scheme above, reporting accuracy, AUC, sensitivity and specificity on each held-out fold.

3.4 Bootstrap-ensemble setting

LightGBM has no continuous weight space to average in the way SWA-Gaussian [5] does for a neural network, so predictive uncertainty is instead obtained by training 30 LightGBM models per fold, each on an independent bootstrap resample (drawn with replacement, same size as the fold's training set) with its own random seed. The 30 members' predicted probabilities are averaged for the point prediction, and their standard deviation, per patient, is reported as that patient's predictive uncertainty – higher values indicate the prediction depended more on which particular resampled model happened to be used, i.e. genuine model disagreement rather than confidence in a single fit.

3.5 Combination study design

The main comparison uses three data types because that is the set MoXGATE's architecture was built for. To ask whether adding the other three (copy number, mutations, protein) helps, LightGBM was also run on every one of the 63 non-empty combinations of all six data types: 6 singles, 15 pairs, 20 triples, 15 quadruples, 6 quintuples and all six together.

Protein was measured for far fewer patients than the other types, so all combinations cannot share one patient set without discarding most of the cohort. Two common groups are used: *group A*, the 392 patients who have all five non-protein types (217 oral, 175 non-oral; 31 combinations), and *group B*, the 256 patients who also have protein (150 oral, 106 non-oral; all 63 combinations). Within a group every combination sees identical patients and identical folds, so comparisons between combinations are paired. This matters because a combination containing protein is

forced onto the smaller group; comparing its accuracy with a non-protein combination on the larger group would mix a cohort effect into the result.

This study builds the label from the ICD-10 site code in its "C14.8 excluded" version: the 70 patients coded C14.8 (overlapping lesion of lip, oral cavity and pharynx, which spans both regions) and the one mandible tumour are dropped; oral is C00 and C02–C06, non-oral is C01, C09, C10, C13 and C32. The main comparison in Section 4.1 instead uses the original anatomic-subdivision label, which counts all 70 C14.8 patients as oral, so accuracy figures in the two parts are not directly comparable.

Preprocessing follows Section 2.3 for each data type separately, using training patients only, except that features are not standardized (tree models are unaffected by rescaling). It depends only on the fold and the data type, so it is computed once per fold and shared by every combination. LightGBM uses the configuration of Section 3.2. Each combination is compared with the better of its own member data types by a paired t-test over the five shared folds. Folds share training data, so this test is optimistic, and with 57 such comparisons about three would reach $p < 0.05$ by chance alone.

4 Results

4.1 Point-estimate LightGBM

Data type	Accuracy	AUC	Sensitivity	Specificity
Gene expression	0.870	0.930	0.921	0.790
DNA methylation	0.927	0.949	0.955	0.882
microRNA	0.851	0.903	0.921	0.742
All three combined	**0.931**	**0.955**	0.955	0.892

Combining all three data types gives the highest accuracy and AUC, but only by 0.004 accuracy over DNA methylation alone (0.931 vs. 0.927), and the bootstrap-ensemble setting below reverses the order (0.927 vs. 0.929). A difference this small, which changes sign between two settings, is no evidence that combining helps. It matches the companion LightGBM report, where combining data types never significantly improved on the best single member, and it matches this project's own MoXGATE results, where the combined model did not beat methylation alone either (0.843 vs. 0.853 for the plain model, 0.895 vs. 0.908 with SWAG). No significance test was run on the LightGBM difference.

4.2 Bootstrap-ensemble LightGBM

Data type	Accuracy	AUC	Mean uncertainty
Gene expression	0.876	0.929	0.0618
DNA methylation	0.929	0.951	**0.0365**
microRNA	0.836	0.903	0.0844
All three combined	0.927	0.954	0.0383

Relative to the point-estimate setting, bootstrap ensembling changes accuracy by +0.006 (gene expression), +0.002 (methylation), −0.015 (microRNA) and −0.004 (combined): no consistent direction, so it neither helps nor hurts overall here. What the ensemble adds is a per-patient uncertainty estimate, and that estimate tracks data-type reliability. Among the single data types, DNA methylation, the most accurate, has the lowest mean predictive uncertainty (0.0365) and microRNA, the least accurate, the highest (0.0844); the combined model (0.0383) sits close to methylation.

4.3 All six data types and all 63 combinations

Table 1 gives each data type alone on both patient groups beside the companion report's LightGBM figures.

Data type	Companion	Group A (392)	Group B (256)
Gene expression	0.876	0.873	0.859
DNA methylation	0.937	**0.941**	**0.918**
microRNA	0.830	0.826	0.840
Copy number	0.651	0.719	0.656
Mutations	0.653	0.612	0.645
Protein	0.724	–	0.707

Table 1: Each data type alone, LightGBM accuracy, C14.8 excluded. Companion: mean over 10 splits on 414 patients (274 for protein). This study: 5-fold cross-validation.

DNA methylation is the strongest single type in both groups, well ahead of gene expression and microRNA, with protein, copy number and mutations far weaker. The ordering and sizes match the companion report closely; the largest differences are copy number on group A (0.719 against 0.651) and mutations (0.612 against 0.653), where the data types are defined differently (Section 2.2).

Tables 2 and 3 give every pair; the diagonal is the single type.

	RNA	Meth	miRNA	CNV	Mut	Prot
RNA	0.859	0.926	0.848	0.859	0.840	0.859
Methylation	0.926	0.918	0.914	0.922	0.918	0.926
miRNA	0.848	0.914	0.840	0.828	0.817	0.824
Copy number	0.859	0.922	0.828	0.656	0.667	0.734
Mutations	0.840	0.918	0.817	0.667	0.645	0.726
Protein	0.859	0.926	0.824	0.734	0.726	0.707

Table 2: Pairwise accuracy, group B (256 patients, all six types).

	RNA	Meth	miRNA	CNV	Mut
RNA	0.873	0.944	0.888	0.870	0.870
Methylation	0.944	0.941	0.939	0.936	0.936
miRNA	0.888	0.939	0.826	0.839	0.837
Copy number	0.870	0.936	0.839	0.719	0.717
Mutations	0.870	0.936	0.837	0.717	0.612

Table 3: Pairwise accuracy, group A (392 patients, five non-protein types).

Adding a second data type to methylation does not help measurably. On group A the four pairs containing methylation score 0.936–0.944 against 0.941 for methylation alone; on group B the five pairs score 0.914–0.926 against 0.918. The best pair in each group (RNA + methylation, 0.944; methylation + protein, 0.926) is ahead by 0.003 and 0.008, about one and two patients respectively.

Using every data type does not help either. On group B all six types reach 0.926 against 0.918 for methylation alone on the same patients and folds (difference +0.008, $p = 0.37$; higher in 3 folds, equal in 1, lower in 1). On group A all five non-protein types reach 0.936 against 0.941 (-0.005, $p = 0.38$; 1 higher, 1 equal, 3 lower). The companion report found +0.005 ($p = 0.19$) for the same comparison. Both differences here are about two patients.

The cohort effect is larger than any effect of adding data types to methylation. Methylation alone scores 0.941 on group A but 0.918 on group B. That 0.023 difference comes only from which patients are included, and it is about three times the largest gain from adding a data type to methylation (+0.008), so accuracy should only be compared within a group.

Across all 57 comparisons of a combination with the better of its members (each judged once, on the largest group that contains its types), 55 show no significant difference. One is significantly better (copy number + mutations + protein, 0.738, against protein alone, 0.707; $p = 0.034$) and one significantly worse (RNA + copy number + mutations + protein, 0.844, against RNA alone, 0.859; $p = 0.016$). With 57 tests about three would reach $p < 0.05$ by chance, so these two are what chance predicts; the companion report's 171 comparisons gave 164 not significant, one better and six worse. The one gain suggests weak data types carry a little complementary information (combining copy number, mutations and protein lifts the best of them by 0.031), but nowhere near enough to matter next to methylation.

Types	Group A (392)	Group B (256)
1	methylation, 0.941	methylation, 0.918
2	RNA + methylation, 0.944	methylation + protein, 0.926
3	RNA + methylation + CNV, 0.944	RNA + methylation + miRNA, 0.926
4	methylation + miRNA + CNV + mutations, 0.936	RNA + methylation + miRNA + CNV, 0.930
5	all five, 0.936	RNA + methylation + miRNA + CNV + protein, 0.930
6	–	all six, 0.926

Table 4: Best combination of each size (accuracy).

Table 4 is essentially flat. The small rise on group B (0.918 to 0.930) should not be read as a benefit: each entry is the maximum over up to 20 noisy estimates, which is biased upward even when no combination is truly better. The paired comparisons above are the unbiased test.

5 Explainability: which genes drive the methylation signal

Section 4 above establishes *that* DNA methylation is the strongest single data type on this task; this section asks *why*, by identifying which specific CpG probes the trained models actually rely on and what genes those probes sit near.

5.1 Method

Two independent rankings of the 5,000 ANOVA-selected methylation probes were computed on the full 477-patient cohort (not per-fold, since this analysis inspects the final trained model rather than estimating accuracy): the one-way ANOVA F-statistic used for feature selection itself, and LightGBM's own feature-importance scores (how often each probe was actually used to split a tree, summed across all 500 trees, after training on the ANOVA-selected feature set). The union of both methods' top 30 probes (54 probes total, reflecting partial overlap between the two rankings) was then mapped to the nearest gene using UCSC Genome Browser's REST API: each probe ID was first resolved to a genomic position via UCSC's `/search` endpoint, then that position was queried against the RefSeq gene annotation track (`refGene`) to identify any overlapping gene.

5.2 Agreement between the two ranking methods

Only 6 of the 30 top-ranked probes (20%) appear in both methods' top-30 lists. This is not the two methods disagreeing about what matters, but a direct consequence of methylation's

correlation structure: nearby CpG sites on the same gene tend to rise and fall together, so a single underlying signal is often carried by many individually-significant, mutually redundant probes. The gene-mapping results below confirm this directly – seven or more of the top-ranked ANOVA probes sit near a single gene, *PITX2*, which ANOVA's per-probe scoring ranks highly for every one of them, while LightGBM's trees need only one representative from that cluster before moving on to a different genomic region. The 6 probes that remain in both top-30 lists despite this redundancy-driven divergence are on stronger footing than either ranking alone would suggest: a signal strong enough to survive both an independent, per-feature statistical test and a model that actively economizes away redundant features.

5.3 Gene mapping results

42 of the 54 probes (78%) resolved to a RefSeq gene; the remaining 12 had no gene overlapping their genomic position, consistent with roughly half of the human genome being intergenic – an expected outcome, not a failure of the lookup. *PITX2* alone accounts for seven or more of the resolved probes – a striking concentration for a single gene, and itself direct evidence of the correlated-probe-cluster effect described above. Several *HOX* cluster genes (*HOXA1*, *HOXA3*, *HOXB1*) also each recur across multiple probes. The remaining resolved probes each map to their own distinct gene, including *TFAP2A*, *FOXC1*, *PAX9* and *LHX5* among others.

5.4 Biological interpretation

PITX2, the recurring *HOXA*/*HOXB* cluster genes, *TFAP2A*, *FOXC1* and *PAX9* are not an arbitrary sample of the genome. All have specific, well-documented roles in craniofacial and pharyngeal arch development: HOX genes are established master regulators of pharyngeal arch identity [9], the embryonic structures that give rise to different regions of the head and neck; *PITX2* and *FOXC1* act in the same craniofacial and pharyngeal-pouch patterning pathway [10] (co-mutated in Axenfeld-Rieger syndrome, which affects craniofacial and dental structures); *TFAP2A* is required for neural crest cell development [11], the embryonic cell population that forms much of the facial and pharyngeal skeleton; and *PAX9* is specifically implicated in palate and tooth development [12].

This is consistent with a specific, checkable explanation for why DNA methylation outperforms gene expression and microRNA throughout this report: the oral cavity and the adjacent pharynx/larynx arise from different embryonic pharyngeal arches, and DNA methylation appears to retain a signature of that developmental origin at genes that established the identity of those arches during embryogenesis, more so than transcriptional data reflecting the tumour's current state. This mirrors an established clinical use of DNA methylation profiling more broadly, where methylation-based classifiers correctly identify a tumour's tissue of origin in roughly 87% of cancer-of-unknown-primary cases [13], precisely because methylation is understood to track developmental lineage more robustly than gene expression does.

5.5 Caveats

This analysis uses the full cohort rather than a held-out split, and therefore describes the final, fully-trained model rather than providing an independent accuracy estimate for the gene list itself; the specific probe ranking should be treated as hypothesis-generating, not as a validated biomarker panel, which would require its own dedicated, prospectively held-out validation. The gene-position mapping also reflects one representative RefSeq transcript per queried position and does not attempt to resolve cases where multiple overlapping genes or isoforms are present at the same coordinate.

6 Comparison with MoXGATE on the identical cohort and folds

An independently reproduced, corrected implementation of MoXGATE [3] (matching the architecture described in the companion improved-MoXGATE report: self-attention [6] across the patients of a batch, a non-residual feed-forward block per modality encoder, per-data-type auxiliary classifier heads, and SWA-Gaussian weight averaging) was evaluated on the identical 477-patient cohort and 5-fold cross-validation split as the LightGBM results above.

Data type	LightGBM (point est.)	MoXGATE (point est.)	MoXGATE (SWAG)
Gene expression	0.870	0.780	0.819
DNA methylation	0.927	0.853	0.908
microRNA	0.851	0.776	0.787
All three combined	0.931	0.843	0.895

The SWAG column is averaged over two independent runs (SWAG draws its 30 weight samples from a random generator that is not seeded, so repeated runs vary slightly by design; the two runs agreed within 0.005 accuracy on every data type except the combined setting, where they differed by 0.004 – a small enough spread that the average is reported as the more stable figure rather than either single run).

LightGBM's point-estimate model outperforms MoXGATE's point-estimate model on every data type, by 0.074 to 0.090. SWA-Gaussian weight averaging closes much of that gap on DNA methylation (0.853 to 0.908), leaving MoXGATE 0.019 behind LightGBM's 0.927, which is comparable to the fold-to-fold standard deviation of the MoXGATE methylation runs (0.014–0.019). For the other settings a clear gap remains: 0.051 on gene expression, 0.064 on microRNA and 0.036 for the combined setting.

7 Label-boundary sensitivity and per-site accuracy

The label used in Sections 4 and 6 counts all 70 C14.8 (overlapping lesion) patients as oral, since that is how the original anatomic-subdivision field records them. A separate ICD-10-based construction allows three versions: C14.8 excluded, C14.8 counted as oral (matching the label used elsewhere in this report), and "clean sites", which also drops the 25 patients coded C01 (base of tongue, anatomically oropharyngeal despite sitting directly behind the oral tongue). Each was evaluated with methylation-only LightGBM, 5-fold cross-validation.

Label version	Patients	Accuracy	Sensitivity	Specificity
C14.8 excluded	457	0.939	0.948	0.927
C14.8 as oral	527	0.932	0.944	0.913
Clean sites	432	0.958	0.960	0.956

Dropping base of tongue in addition to C14.8 raises accuracy by 0.026 over C14.8-excluded and 0.019 over C14.8-as-oral. Breaking the C14.8-excluded run's out-of-sample predictions down by individual ICD-10 code shows why:

Code	Site	Patients	Accuracy
C01	base of tongue	25	0.680
C10.9	oropharynx	11	0.727
C09.9	tonsil	42	0.929
C02.9	tongue (oral part)	131	0.939
C04.9	floor of mouth	55	0.945
C06.0	mouth, other	19	0.947
C06.9	mouth, other	23	0.957
C32.9	larynx	118	0.992

Larynx, anatomically the most distinct site from the oral cavity, is almost error-free (0.992). The three oropharyngeal or oropharynx-adjacent sites – base of tongue, oropharynx and tonsil – make up only 17% of these 457 patients but account for half (14 of 28) of the errors; base of tongue alone, easily confused with the oral tongue it sits directly behind, is the single hardest site (0.680). This points at the label boundary, not the model, as the limiting factor: unambiguous sites are classified almost perfectly, and errors concentrate exactly where the underlying anatomy is itself ambiguous.

8 Status of further checks

Three further checks were built and verified on synthetic data, matching this report's other real-data scripts, but have not yet produced results on the real cohort reported here. An HPV/p16 confound check ran once using the wrong clinical field (viral testing by ISH, 89 patients with a value) rather than the intended one (p16 immunohistochemistry, 115 patients); the codebase now corrects this, and that run's numbers are not reported here pending a corrected rerun. A hospital-confound check, which splits cross-validation folds so no submitting institution appears in both the training and test set, is implemented but unrun. Whether MoXGATE's accuracy degrades as weak data types are added – the pattern the companion MoXGATE report found for its own network – remains untested here; the combination study of Section 4.3 covers LightGBM only.

9 Discussion

Three findings are worth setting against the companion LightGBM report's own conclusions, since this evaluation used an independently built pipeline on an overlapping but not identical setup (three data types in the main comparison and six in the combination study here, six there; 5-fold cross-validation here against ten random 72/8/20 splits there).

First, DNA methylation's dominance as the strongest single data type reproduces cleanly: it is the strongest single data type in every setting reported here, matching the companion report's own central finding.

Second, combining data types does not help, matching the companion report, and the combination study now tests this directly with all six data types. On the same patients and folds, all six types differ from methylation alone by +0.008 (group B) and all five non-protein types by −0.005 (group A), about two patients either way; 55 of 57 combinations show no significant difference from their best member, and the two that do are what chance predicts at that number of tests. In the three-type comparison the combination differs from methylation alone by +0.004 (point estimate) and −0.002 (ensemble), and for MoXGATE by −0.010 and −0.013 (without and with SWAG). The one thing this study cannot say is whether extra data types damage the *network*, as the companion report found; MoXGATE was run on three data types only.

Third, gradient-boosted trees' advantage over the multi-omic attention network narrows once the network is evaluated with SWA-Gaussian weight averaging, but does not disappear. On

methylation the gap falls from 0.074 to 0.019, comparable to the fold-to-fold spread; on the other settings it remains between 0.036 and 0.064. This is consistent with the improved-MoXGATE report's stated motivation for SWAG, that a single checkpoint chosen by early stopping on a small validation split is an unreliable point to trust, but on this cohort averaging over weight samples recovers only part of the gap to gradient boosting.

References

[1] Improved MoXGATE for Oral versus Non-Oral Classification of Head and Neck Squamous Cell Carcinoma from Multi-Omic TCGA Data. Unpublished companion technical report, 2026.

[2] LightGBM baseline for oral versus non-oral classification of TCGA-HNSC from multi-omic data: an exhaustive combination and label-definition study. Unpublished companion technical report, 2026.

[3] S. Dip et al. MoXGATE: Modality-Aware Cross-Attention for Multi-Omic Gastrointestinal Cancer Subtype Classification. In *ICLR 2025 Workshop on AI for Nucleic Acids (AI4NA)*, 2025. arXiv:2506.06980.

[4] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

[5] W. J. Maddox, T. Garipov, P. Izmailov, D. Vetrov, and A. G. Wilson. A Simple Baseline for Bayesian Uncertainty in Deep Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

[6] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention Is All You Need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.

[7] The Cancer Genome Atlas Network. Comprehensive genomic characterization of head and neck squamous cell carcinomas. *Nature*, 517:576–582, 2015. doi:10.1038/nature14129.

[8] M. J. Goldman, B. Craft, M. Hastie, K. Repečka, F. McDade, A. Kamath, A. Banerjee, Y. Luo, D. Rogers, A. N. Brooks, J. Zhu, and D. Haussler. Visualizing and interpreting cancer genomics data via the Xena platform. *Nature Biotechnology*, 38:675–678, 2020. doi:10.1038/s41587-020-0546-8.

[9] A. Iulianella and P. A. Trainor. Hox gene control of neural crest cell, pharyngeal arch and craniofacial patterning. In *Craniofacial Development*, Current Topics in Developmental Biology. Elsevier, 2014.

[10] P. Dressler, N. Weissschuh, F. Schuettauf, C. Wolf, B. Wissinger, and E. Gramer. Dental and craniofacial anomalies associated with Axenfeld-Rieger syndrome with PITX2 mutation. *Case Reports in Medicine*, 2010:621984, 2010. doi:10.1155/2010/621984.

[11] H. Schorle, P. Meier, M. Buchert, R. Jaenisch, and P. J. Mitchell. Transcription factor AP-2 essential for cranial closure and craniofacial development. *Nature*, 381:235–238, 1996. doi:10.1038/381235a0.

[12] O. Bonczek, V. J. Balcar, and O. Šerý. PAX9 gene mutations and tooth agenesis: a review. *Clinical Genetics*, 92(5):467–476, 2017. doi:10.1111/cge.12986.

[13] S. Moran, A. Martínez-Cardús, S. Sayols, et al. Epigenetic profiling to classify cancer of unknown primary: a multicentre, retrospective analysis. *Lancet Oncology*, 17(10):1386–1395, 2016. doi:10.1016/S1470-2045(16)30297-2.